CS24353 • MASTER REVISION

Software Engineering (CS24353) 10-Page Master Revision

Universal Formulas, McCabe Cyclomatic Complexity, COCOMO Numericals, FP Analysis & Solved Flashcards

Author: Shaswat Raj (BIT Mesra) **Academic Year:** 2024–25 (5th Sem CSE) **Standard:** IEEE & ISO/IEC Standards

Master Formula Compendium & Process Model Cheat-Sheets

🏛 Master Flashcard 1: Universal Estimation Formulas

- **Cyclomatic Complexity:**

$$V(G) = E - N + 2P = P + 1 = \text{Enclosed Regions} + 1$$

- **Basic COCOMO Effort & Time:**

$$E = a(\text{KLOC})^b \quad (\text{Organic: } 2.4, 1.05;\ \text{Semi: } 3.0, 1.12;\ \text{Embedded: } 3.6, 1.20)$$

$$T_{\text{dev}} = c(E)^d \quad (\text{Organic: } 2.5, 0.38;\ \text{Semi: } 2.5, 0.35;\ \text{Embedded: } 2.5, 0.32)$$

- **Function Points:**

$$\text{FP} = \text{UFP} \times (0.65 + 0.01 \times \text{TDI})$$

- **Availability:**

$$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}}$$

🏛 Master Flashcard 2: Cohesion vs Coupling Hierarchy

Ranking	Cohesion (Internal Strength - High is Good)	Coupling (Interdependence - Low is Good)
Best (10/10)	Functional Cohesion	Data / Message Coupling
Very Good	Sequential Cohesion	Stamp (Data-Structure) Coupling
Good	Communicational Cohesion	Control Coupling (Passing flags)
Fair	Procedural Cohesion	Common Coupling (Global variables)
Worst (0/10)	Coincidental Cohesion	Content Coupling (Direct internal mutation)

🏛 Master Flashcard 3: CMMI 5 Maturity Levels

$$\text{1. Initial (Ad-hoc)} \rightarrow \text{2. Managed (Project)} \rightarrow \text{3. Defined (Org-wide)} \rightarrow \text{4. Quantitatively Managed (Stats)} \rightarrow \text{5. Optimizing (Continuous)}$$

🏛 Master Flashcard 4: Software Maintenance Categories

$$\text{Corrective (Fix bugs: 20\%)} \quad \text{Adaptive (New OS/DB: 25\%)} \quad \text{Perfective (New features: 50\%)} \quad \text{Preventive (Refactor: 5\%)}$$

Master Examination Problem Cheatsheets & Flashcards

🏛 Master Flashcard 5: Complete Process Models Decision Matrix

Project Profile	Optimal Process Model	Engineering Justification
Flight Control / Pacemaker	Waterfall / V-Model	Strict formal verification, unchanging critical specifications.
Novel Startup Mobile App	Agile Scrum / XP	Fluid requirements, weekly user feedback, rapid MVP iterations.
Large Multi-Million Enterprise ERP	Boehm's Spiral	Massive technological and financial risks evaluated per cycle.
Small Web Tool with Reusable GUI	RAD Model	4GL component assembly delivers shippable tool in 60 days.

🏛 Master Flashcard 6: Complete Testing Coverage Hierarchy

$$\text{Statement } (C_0) \subset \text{Branch/Decision } (C_1) \subset \text{Condition Coverage} \subset \text{MC/DC} \subset \text{Multiple Condition Coverage}$$

🏛 Master Flashcard 7: COCOMO Equations & Multipliers

$$\text{Organic: } E = 2.4(\text{KLOC})^{1.05},\ T_{\text{dev}} = 2.5(E)^{0.38}$$
$$\text{Semi-Detached: } E = 3.0(\text{KLOC})^{1.12},\ T_{\text{dev}} = 2.5(E)^{0.35}$$
$$\text{Embedded: } E = 3.6(\text{KLOC})^{1.20},\ T_{\text{dev}} = 2.5(E)^{0.32}$$

🏛 Master Flashcard 8: Function Points Value Adjustment Factor

$$\text{VAF} = 0.65 + 0.01 \times \sum_{i=1}^{14} C_i \qquad \text{Adjusted FP} = \text{UFP} \times \text{VAF}$$

🏛 Master Flashcard 9: Complete Software Metric Formulas

- **Halstead Volume:** $V = (N_1 + N_2)\log_2(n_1 + n_2)$
- **Halstead Difficulty:** $D = \frac{n_1}{2} \times \frac{N_2}{n_2}$
- **Halstead Effort:** $E = D \times V$
- **Halstead Time:** $T = E/18$ seconds

🏛 Master Flashcard 10: Complete SCM Baseline Lifecycle

$$\text{Working Copy} \xrightarrow{\text{Commit}} \text{Repository} \xrightarrow{\text{Review/Audit}} \text{Approved Baseline} \xrightarrow{\text{CR/CCB}} \text{New Version}$$

Master Examination Flashcard Compendium

🏛 Master Flashcard 11: Formal Inspections vs Walkthroughs

Dimension	Formal Inspection (Fagan)	Walkthrough
Formality	Highly formal 6-step process with formal checklists.	Informal peer review session.
Leadership	Led by trained impartial **Moderator**.	Led by the author of the code/document.
Goal	Defect discovery & metrics logging (no solutions discussed).	Knowledge sharing, educational review.

🏛 Master Flashcard 12: Testing Coverage Metrics Summary

$$\text{Cyclomatic Complexity: } V(G) = E - N + 2P = P + 1 = \text{Regions} + 1$$
$$\text{Halstead Volume: } V = (N_1 + N_2)\log_2(n_1 + n_2) \qquad \text{Effort: } E = \left(\frac{n_1 N_2}{2n_2}\right) \times V$$

🏛 Master Flashcard 13: Software Maintenance Distribution

$$\text{Perfective (50\%: New features)} \gg \text{Adaptive (25\%: OS/DB)} \gg \text{Corrective (20\%: Bug fixes)} \gg \text{Preventive (5\%: Refactoring)}$$

🏛 Master Flashcard 14: Comprehensive SE Estimation Cheat-Sheet

$$\text{COCOMO: } E = a(\text{KLOC})^b \times \text{EAF} \qquad \text{FP: UFP} \times (0.65 + 0.01 \times \text{TDI})$$
$$\text{Halstead: } V = N\log_2 n \qquad D = \frac{n_1 N_2}{2n_2} \qquad E = D \times V \qquad T = E/18$$

🏛 Master Flashcard 15: Full Testing Hierarchy Cheat-Sheet

$$\text{Unit (JUnit)} \to \text{Integration (Stubs/Drivers)} \to \text{System (Security/Perf)} \to \text{UAT (Acceptance)}$$

Comprehensive 10-Page Master Examination Compendium

🏛 Master Flashcard 16: Complete Process Models Comparison

Model	Key Feature	Risk Handling	Best When
Waterfall	Sequential milestones	Zero explicit risk	Stable, fully known specs
Prototyping	Mockups for discovery	Requirement risk	Unclear user needs
Spiral	Risk-driven quadrants	Exhaustive analysis	Large, high-risk systems
Agile Scrum	2-4 week sprints	Continuous mitigation	Rapidly evolving markets

🏛 Master Flashcard 17: Complete Estimation Formulas Summary

$$\text{Cyclomatic Complexity: } V(G) = E - N + 2P = P + 1 = \text{Regions} + 1$$

$$\text{Basic COCOMO Effort: } E = a(\text{KLOC})^b \qquad \text{Development Time: } T_{\text{dev}} = c(E)^d$$

$$\text{Function Points: FP} = \text{UFP} \times (0.65 + 0.01 \times \text{TDI})$$

$$\text{Halstead Effort: } E = \left(\frac{n_1 N_2}{2n_2}\right) \times (N_1 + N_2)\log_2(n_1 + n_2)$$

🏛 Master Flashcard 18: Quality Assurance & Standards Summary

$$\text{CMMI Levels: 1. Initial} \rightarrow \text{2. Managed} \rightarrow \text{3. Defined} \rightarrow \text{4. Quantitatively Managed} \rightarrow \text{5. Optimizing}$$

🏛 Master Flashcard 19: Complete Examination Metric Compendium

$$\text{COCOMO: } E = a(\text{KLOC})^b \times \text{EAF} \qquad \text{Time: } T = c(E)^d \qquad \text{Staff: } N = E/T$$

$$\text{Function Points: FP} = \text{UFP} \times (0.65 + 0.01 \times \text{TDI}) \qquad \text{Cyclomatic Complexity: } V(G) = E - N + 2P$$

$$\text{Availability: } A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} \qquad \text{Halstead Volume: } V = (N_1 + N_2)\log_2(n_1 + n_2)$$

🏛 Master Flashcard 20: Software Engineering Golden Rules

Maximize Cohesion • Minimize Coupling • Write Tests First • Manage Technical Debt • Baselined SCM!

🏛 Master Flashcard 21: Earned Value Management Summary

$$\text{CV} = \text{EV} - \text{AC} \qquad \text{SV} = \text{EV} - \text{PV} \qquad \text{CPI} = \text{EV}/\text{AC} \qquad \text{SPI} = \text{EV}/\text{PV}$$

🏛 Master Flashcard 22: Quality & Defect Metrics Summary

$$\text{Defect Density} = \frac{\text{Defects}}{\text{KLOC}} \qquad \text{DRE} = \frac{\text{Internal Defects}}{\text{Internal} + \text{External}} \times 100\%$$

🏛 Master Flashcard 23: Complete Software Metrics Summary

$$\text{Maintainability Index: MI} = 171 - 5.2\ln(V) - 0.23V(G) - 16.2\ln(\text{LOC})$$

$$\text{Defect Removal Efficiency: DRE} = \frac{E}{E + D} \times 100\% \qquad \text{Availability: } A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

🏛 Master Flashcard 24: Final Master Examination Review Compendium

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🏛 Master Flashcard 25: Complete Architectural Design Patterns

- **Creational:** Singleton, Factory, Builder, Prototype.
- **Structural:** Adapter, Decorator, Facade, Proxy.
- **Behavioral:** Observer, Strategy, Command, State, Template Method.

🏛 Master Flashcard 26: Complete Testing & Reliability Metrics

$$\text{Cyclomatic Complexity: } V(G) = E - N + 2P \qquad \text{Availability: } A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

$$\text{Halstead Effort: } E = D \times V \qquad \text{COCOMO: } E = a(\text{KLOC})^b \times \text{EAF}$$

🏛 Master Flashcard 27: Complete Estimation Formulas Summary

$$\text{COCOMO: } E = a(\text{KLOC})^b \times \text{EAF} \qquad T = c(E)^d \qquad N = E/T$$

$$\text{Function Points: FP} = \text{UFP} \times (0.65 + 0.01 \times \text{TDI}) \qquad \text{Complexity: } V(G) = E - N + 2P$$

$$\text{Availability: } A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} \qquad \text{Halstead Volume: } V = (N_1 + N_2)\log_2(n_1 + n_2)$$

🏛 Master Flashcard 28: Software Engineering 100% Mastery Checklist

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🏛 Master Flashcard 29: Complete Software Engineering Life-Cycle Summary

Requirements (IEEE 830) $\rightarrow$ Design (UML/SOLID) $\rightarrow$ Testing (MC/DC/BVA) $\rightarrow$ Estimation (COCOMO/FP) $\rightarrow$ Maintenance (SCM/CMMI)

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🏛 Master Flashcard 31: Universal Software Engineering Examination Summary

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🏛 Master Flashcard 32: Software Metrics Master Cheatsheet

$$\text{Halstead Volume: } V = (N_1 + N_2)\log_2(n_1 + n_2) \qquad \text{Effort: } E = \left(\frac{n_1 N_2}{2n_2}\right) \times V$$

$$\text{COCOMO Effort: } E = a(\text{KLOC})^b \times \text{EAF} \qquad \text{Development Time: } T_{\text{dev}} = c(E)^d$$

$$\text{Function Points: FP} = \text{UFP} \times (0.65 + 0.01 \times \text{TDI})$$

$$\text{Cyclomatic Complexity: } V(G) = E - N + 2P = P + 1 = \text{Regions} + 1$$

$$\text{Availability: } A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}}$$

🏛 Master Flashcard 33: Complete Software Engineering Formulas Compendium

$$\text{Halstead: } V = (N_1 + N_2)\log_2(n_1 + n_2) \qquad D = \frac{n_1 N_2}{2n_2} \qquad E = D \times V \qquad T = E/18$$

$$\text{COCOMO: } E = a(\text{KLOC})^b \times \text{EAF} \qquad T_{\text{dev}} = c(E)^d \qquad N = E/T_{\text{dev}}$$

$$\text{Function Points: FP} = \text{UFP} \times (0.65 + 0.01 \times \text{TDI}) \qquad \text{Complexity: } V(G) = E - N + 2P$$

$$\text{Availability: } A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} \qquad \text{DRE: } \frac{E}{E + D} \times 100\%$$

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🏛 Master Flashcard 36: Final Universal Formula Sheet

$$\text{Halstead: } V = (N_1 + N_2)\log_2(n_1 + n_2) \qquad E = \left(\frac{n_1 N_2}{2n_2}\right) \times V \qquad T = E/18$$

$$\text{COCOMO: } E = a(\text{KLOC})^b \times \text{EAF} \qquad T = c(E)^d \qquad N = E/T$$

$$\text{Function Points: FP} = \text{UFP} \times (0.65 + 0.01 \times \text{TDI})$$

$$\text{Cyclomatic Complexity: } V(G) = E - N + 2P = P + 1 = \text{Regions} + 1$$

$$\text{Availability: } A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}}$$

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🏛 Master Flashcard 39: Complete Metrics Cheat Sheet

$$\text{Halstead: } V = N\log_2 n \qquad E = D \times V \qquad T = E/18$$

$$\text{COCOMO: } E = a(\text{KLOC})^b \times \text{EAF} \qquad T = c(E)^d$$

$$\text{Function Points: FP} = \text{UFP} \times (0.65 + 0.01 \times \text{TDI})$$

$$\text{Cyclomatic Complexity: } V(G) = E - N + 2P = P + 1 = \text{Regions} + 1$$

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🏛 Master Flashcard 42: Complete Software Engineering Golden Rules

Maximize Cohesion • Minimize Coupling • Write Tests First • Manage Technical Debt • Strict SCM Baselines!

Comprehensive 10-Page Master Examination Compendium

🏛 Master Flashcard 43: Complete SDLC Models Selection Taxonomy

Project Type	Optimal Model	Primary Technical Justification
Safety-Critical Avionics	V-Model / Formal Methods	Mandatory 100% verification traceability for FAA DO-178C.
High-Innovation Web App	Agile Scrum / XP	Continuous user feedback loops and dynamic sprint backlog re-prioritization.
Multi-Million Dollar ERP	Boehm's Spiral	Explicit risk assessment and prototyping prior to capital expenditure.
Internal Corporate Tool	RAD Model	Rapid component assembly and visual 4GL GUI builders.

🏛 Master Flashcard 44: Complete Cost Estimation Equations

$$\text{COCOMO Effort: } \mathbf{E} = \mathbf{a}(\text{KLOC})^{\mathbf{b}} \times \prod_{\mathbf{i}=\mathbf{1}}^{\mathbf{15}} \text{EM}_{\mathbf{i}} \qquad \text{Development Time: } \mathbf{T}_{\text{dev}} = \mathbf{c}(\mathbf{E})^{\mathbf{d}}$$

$$\text{Function Points: FP} = \text{UFP} \times \left(\mathbf{0.65} + \mathbf{0.01} \times \sum_{\mathbf{i}=\mathbf{1}}^{\mathbf{14}} \mathbf{C}_{\mathbf{i}}\right)$$

$$\text{Halstead Science: } \mathbf{V} = (\mathbf{N_1} + \mathbf{N_2}) \log_{\mathbf{2}}(\mathbf{n_1} + \mathbf{n_2}) \qquad \mathbf{E} = \left(\frac{\mathbf{n_1 N_2}}{\mathbf{2n_2}}\right) \times \mathbf{V} \qquad \mathbf{T} = \frac{\mathbf{E}}{\mathbf{18}}\ \text{sec}$$

🏛 Master Flashcard 45: Complete Software Testing & Coverage Hierarchy

$$\text{Statement } (\mathbf{C_0}) \subset \text{Branch } (\mathbf{C_1}) \subset \text{Condition Coverage} \subset \text{MC/DC (DO-178C)} \subset \text{Multiple Condition Coverage}$$

$$\text{McCabe Cyclomatic Complexity: } \mathbf{V}(\mathbf{G}) = \mathbf{E} - \mathbf{N} + \mathbf{2P} = \mathbf{P}_{\text{nodes}} + \mathbf{1} = \text{Enclosed Regions} + \mathbf{1}$$

🏛 Master Flashcard 46: Complete Software Reliability & Quality Metrics

$$\text{MTBF} = \text{MTTF} + \text{MTTR} \qquad \text{Availability } \mathbf{A} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} \qquad \text{DRE} = \frac{\mathbf{E}}{\mathbf{E} + \mathbf{D}} \times \mathbf{100\%}$$

$$\text{Defect Density} = \frac{\text{Total Defects}}{\text{KLOC}} \qquad \text{Maintainability Index: MI} = \mathbf{171} - \mathbf{5.2}\ln(\mathbf{V}) - \mathbf{0.23V}(\mathbf{G}) - \mathbf{16.2}\ln(\text{LOC})$$

🏛 Master Flashcard 47: Software Configuration Management (SCM) Lifecycle

$$\text{Working Copy} \xrightarrow{\text{Commit}} \text{Repository} \xrightarrow{\text{Audit/Signoff}} \text{Baseline} \xrightarrow{\text{CR}} \text{CCB Review} \xrightarrow{\text{Approved}} \text{Next Baseline}$$

🏛 Master Flashcard 48: Software Maintenance & Evolution (Lehman's Laws)

$$\text{Maintenance: Perfective } (50\%) > \text{Adaptive } (25\%) > \text{Corrective } (20\%) > \text{Preventive } (5\%)$$

$$\text{Lehman's Laws: I. Continuing Change} \bullet \text{II. Increasing Complexity} \bullet \text{VI. Continuing Growth} \bullet \text{VII. Declining Quality}$$